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PROJECT PHASE DOCUMENT

FITTRACK AI

Personalized Fitness Recommendations Powered by AI

PHASE 1

BRAINSTORMING & IDEATION

Project Name: AI FitTrack

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1.1 Project Title

FitTrack AI – Personalized Fitness Recommendations Powered by AI

Project Name: AI FitTrack

AI FitTrack is an AI-powered fitness tracking backend that enables users to maintain workout records and receive personalized fitness recommendations. It combines workout management capabilities with artificial intelligence to provide useful guidance based on individual fitness information and workout statistics.

1.2 Background of the Project

The initial concept for AI FitTrack emerged from the challenges people may experience when maintaining fitness records manually. Although workouts can be recorded in notebooks or spreadsheets, keeping information organized over an extended period can become inconvenient. Reviewing previous workout details, monitoring progress, and obtaining guidance aligned with personal fitness goals may also require considerable effort.

The proposed application addresses these challenges by bringing workout record management, user authentication, search functionality, and AI-based fitness guidance together within a single backend system. This centralized approach is intended to make fitness information easier to maintain and access.

1.3 Problem Identification

During the brainstorming and ideation stage, the following limitations of conventional manual fitness tracking were identified:

- Workout information can become difficult to organize when it is maintained manually.
- Manual data entry and calculations may introduce inaccuracies into workout records.
- Traditional tracking methods may not provide secure, account-based access to personal fitness information.
- Searching for a specific workout among many records can be time-consuming.

- Users may not receive recommendations tailored to their fitness goals and experience levels.
- Accumulated workout records are not always converted into meaningful fitness insights.
- Manually calculating workout duration, calories burned, and other performance details requires additional effort.

1.4 Proposed Idea

To address the identified limitations, AI FitTrack was proposed as a centralized fitness-management backend. Users can register and securely access their accounts before managing their workout information.

The application exposes RESTful APIs through which authenticated users can create, view, update, delete, and search workout records. It also integrates with Google Gemini AI to generate personalized recommendations and fitness insights using information supplied by the user and relevant workout statistics.

1.5 AI-Based Concept

Artificial intelligence is included to extend the system beyond conventional workout record management. The aim is to provide fitness guidance that is more relevant to each user's circumstances.

For personalized recommendations, the system considers inputs such as the user's age, fitness goal, and experience level. Based on these details, the AI may generate:

- A personalized workout plan.
- Weekly workout suggestions.
- Recommendations appropriate to the user's experience level.
- Motivational guidance.
- General fitness and safety tips.

The system can also generate fitness insights from workout statistics, including the total number of workouts, average workout duration, and total calories burned. These insights are intended to help users understand their recorded activity more clearly. AI-generated guidance is informational and should not be treated as a substitute for professional medical advice.

1.6 Main Objectives

The main objectives established during the brainstorming stage are:

1. To provide a centralized system for maintaining workout records.
2. To support secure access through user authentication.
3. To enable users to manage and search their workout history efficiently.
4. To provide personalized fitness recommendations using Google Gemini AI.
5. To generate useful fitness insights from workout statistics.
6. To develop the backend using a modular structure that can accommodate additional fitness-related features in the future.

1.7 Expected Outcome

AI FitTrack is expected to reduce the effort involved in manual workout tracking and provide users with a secure, organized way to manage their fitness information. By combining workout management with AI-generated recommendations and fitness insights, the system is intended to offer more personalized support to users working towards their fitness goals.

1.8 Phase Summary

The brainstorming and ideation phase established the purpose and core concept of FitTrack AI – Personalized Fitness Recommendations Powered by AI. The key problems associated with manual fitness tracking were identified, and a backend solution combining secure user access, workout management, search capabilities, and Google Gemini AI integration was proposed. These initial objectives provide a foundation for the subsequent phases of project development.